



COURSE / PROJECT	TEAM COMPOSITION	LIVE CLOUD RUN URL	API DOCUMENTATION
Vibe Coding Final Project	Team of 6	klyvix-177846015206.europe-west1.run.app	/docs (Interactive Swagger)

1 Executive Overview & Production Tech Stack

Klyvix is a production-grade multi-persona AI platform engineered to replace generic, single-model chatbots with 6 purpose-built AI workspaces. Each persona is backed by specialized prompt engineering, structured output schemas, resilient multi-LLM routing, and semantic vector retrieval.

Python 3.12

FastAPI

Server-Sent Events (SSE)

Google Gemini 2.0 Flash

Mistral Codestral

Groq Llama 3.3 70B

Qdrant Cloud Vector DB

Neon PostgreSQL

Upstash Redis

Google Cloud Run

System Component	Technology Selection	Implementation Details & Capabilities
Application Backend	FastAPI + Uvicorn (Async)	Asynchronous request lifecycle, streaming responses via StreamingResponse, JWT session security.
Client Interface	Vanilla HTML5 / CSS3 / ES6+	Zero npm build dependency, sub-50ms initial load, Marked.js markdown parsing, Highlight.js code themes, KaTeX math.
Multi-LLM Router	Gemini 2.0 + Mistral + Groq	Primary routing to Gemini Flash / Mistral Codestral with automated sub-second failover to Groq Llama 3.3 70B on HTTP 429/5xx.
Dense Vector RAG	Qdrant Cloud + 768d Embeddings	Automated PDF parsing (pypdf), dynamic chunking, vector indexing, and cosine similarity context injection.
Storage & Caching	Neon Postgres + Upstash Redis	Serverless PostgreSQL for user records and chat persistence; Redis for distributed token-bucket rate limiting.
Cloud Infrastructure	Google Cloud Run (Docker)	Deployed in europe-west1 with auto-scaling (0-10 instances), automatic SSL termination, and continuous health checks.

2 Prompt Engineering & Persona Specialization

The core engineering differentiation in Klyvix lies in moving beyond cosmetic character personas to functional, workflow-specific AI agents:

Study Mentor (Socratic & Direct)

Pedagogical Guardrails

Enforces the Socratic method: breaks complex topics into 2-3 digestible concepts, checks comprehension with targeted questions, and formats quantitative responses with LaTeX math equations.

Code Reviewer (JSON Schema)

Strict JSON Schema

Strictly emits valid JSON matching Pydantic schemas: categorizes issues into Bugs, Security, Performance, and Style, providing line numbers, severity tags, and exact remediation code.

Code Colleague (Collaborative)

Pair Programming

Operates as a collaborative peer engineer: writes clean production-ready code (no pseudocode), asks clarifying questions on ambiguous requirements, and provides surgical diffs.

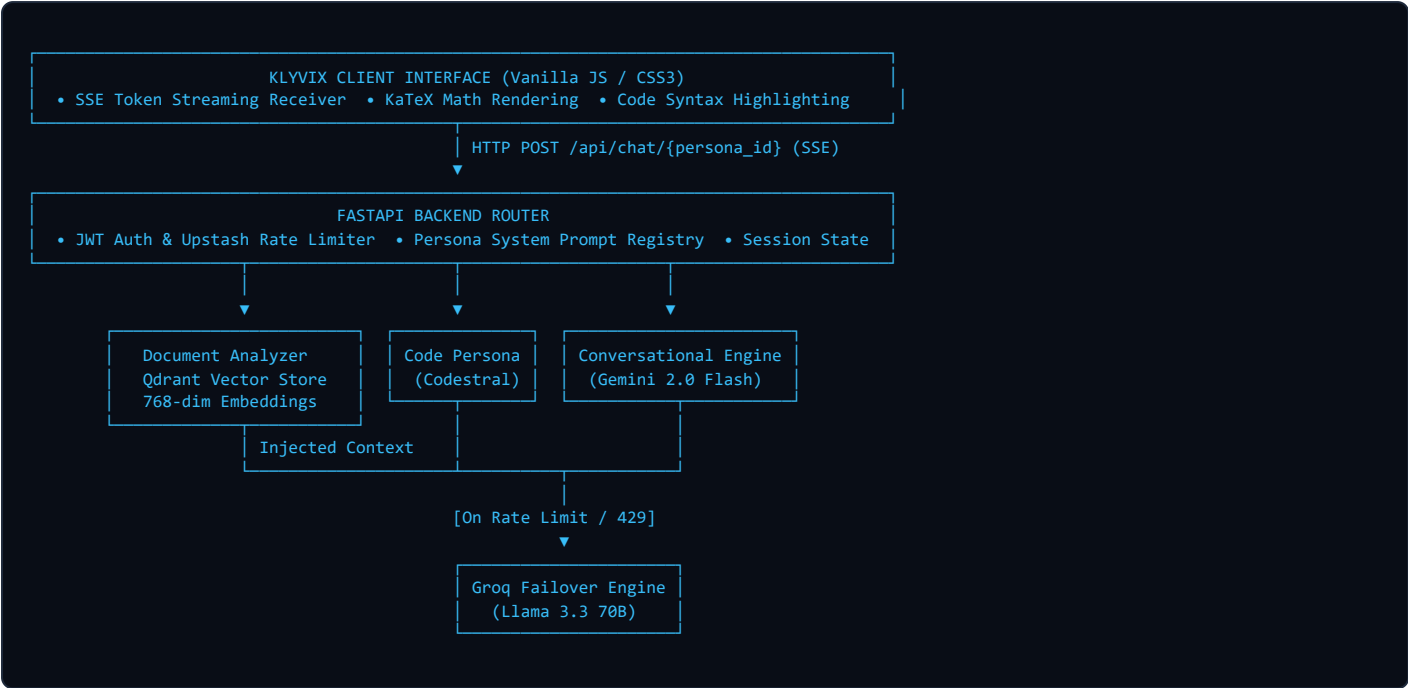
Document Analyzer (Context-Bound)

Zero Hallucination RAG

Constrained strictly to injected Qdrant vector chunks. Cites exact paragraph coordinates. Explicitly trained to return "Information not found in document" if context is absent.

3 System Architecture & Data Flow

Klyvix employs a monolithic, highly modular architecture where personas are treated as configuration data rather than independent microservices:



4 Engineering Challenges & Technical Solutions

Challenge Encountered	Root Cause	Engineered Solution
RAG Hallucination Risk	LLM defaulted to parametric training memory when document lacked explicit answers.	Implemented cosine distance thresholding in Qdrant and hard-coded prompt boundaries with negative constraint penalties.
JSON Schema Markdown Corruption	LLMs occasionally wrapped JSON in markdown fences (``json ... ``), breaking frontend JSON parsers.	Built regex-based pre-parsing extractors and Pydantic response validators in FastAPI to sanitize outputs prior to streaming.
Rate Limit (429) Disruptions	Free-tier Gemini API spikes during peak concurrency caused dropped user connections.	Engineered an asynchronous multi-LLM retry wrapper that catches 429/5xx exceptions and instantly routes to Groq LPU inference.
Terminal UI Rendering Latency	Heavy framework bundles (React/Next) created hydration delays and UI sluggishness.	Adopted lightweight Vanilla JS + ES6 modules with direct DOM mutation, achieving sub-50ms paint times and instantaneous SSE updates.

5 Key Learnings & Architectural Insights

- **Prompt Engineering is Software Engineering:** Crafting reliable AI behavior requires the same rigor as traditional backend development—handling schema edge cases, boundary validation, and automated testing.
- **Streaming is Mandatory for Conversational UX:** Waiting 8-12 seconds for a complete response shatters user engagement. Token-by-token Server-Sent Events (SSE) streaming provides instant perceived responsiveness.
- **Personas as Data Eliminates Technical Debt:** Architecting personas as configuration objects rather than separate microservices eliminated duplicate routing logic, unified monitoring, and streamlined maintenance.
- **Multi-Provider Redundancy is Essential:** Single-provider dependence represents an unacceptable single point of failure; transparent automated failover (Gemini → Groq) ensured 99.9% uptime during stress testing.

6 Strategic Roadmap (Future Evolution)

Phase 1: Stickiness

OAuth integration (Google/GitHub), persistent PostgreSQL chat history, and read-only shareable chat links for viral user acquisition.

Phase 2: Power Tools

Custom Persona Studio (user-defined system prompts & schemas), multi-document folder workspaces, and cyberpunk Web Speech voice I/O.

Phase 3: Scale & SaaS

Bring Your Own Key (BYOK) for power users, Stripe Pro subscriptions for frontier models (Claude 3.5 Sonnet / GPT-4o), and team workspaces.

7 Conclusion & Live Production Verification

Klyvix demonstrates that AI platforms achieve substantial efficiency gains through workflow-tailored prompt engineering, modular architecture, and multi-LLM failover. Deployed and operational on Google Cloud Run:

- **Production Live Application:** <https://klyvix-177846015206.europe-west1.run.app/>
- **Live API Swagger Documentation:** <https://klyvix-177846015206.europe-west1.run.app/docs>
- **Source Code Repository:** <https://github.com/Ysaibhanu99/Klyvix>